

Network Management and Administration (IE2110)
2nd Year, 2nd Semester

Assignment

Zimbra

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Sri Lanka Institute of Information Technology

IT24100816

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Bachelor of Science Special Honors Degree in Information Technology

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Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

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Introduction

This assignment required the deployment and validation of a Linux-based network environment capable of delivering several core administrative services within a controlled virtual laboratory. The completed environment covers the full set of required functions: initial virtual machine preparation, network configuration, dynamic IP address assignment, electronic mail services, centralized directory and authentication services, and web publishing. The implementation used the assignment-specific values of network 192.168.16.0/24, DHCP range 192.168.16.100 - 192.168.16.200, mail domain mail.it16.local, and Active Directory domain it16.local.

The assignment brief specifies the use of three virtual machines: a CentOS server and two Ubuntu client machines. In this implementation, the CentOS server was configured as the primary service host, running the DHCP server, Zimbra Collaboration Suite, and Samba Active Directory Domain Controller. The two Ubuntu client machines were used to verify DHCP IP assignment, domain authentication, and web server access. This consolidated service approach on CentOS ensures all required services are centrally managed while maintaining full compatibility with the assignment requirements.

Based on the captured evidence presented in this report, the environment was tested successfully for DHCP lease allocation (with Ubuntu clients receiving IP addresses in the configured range), Zimbra mail transmission between multiple user accounts on the mail.it16.local domain, Active Directory provisioning with three Organizational Units (Admins, Staff, Students) and nine user accounts, Ubuntu client domain authentication, and client-side access to the required web page displaying "HELLO NMA" with a blue background.

1. Setup

1.1 Virtual machine creation.

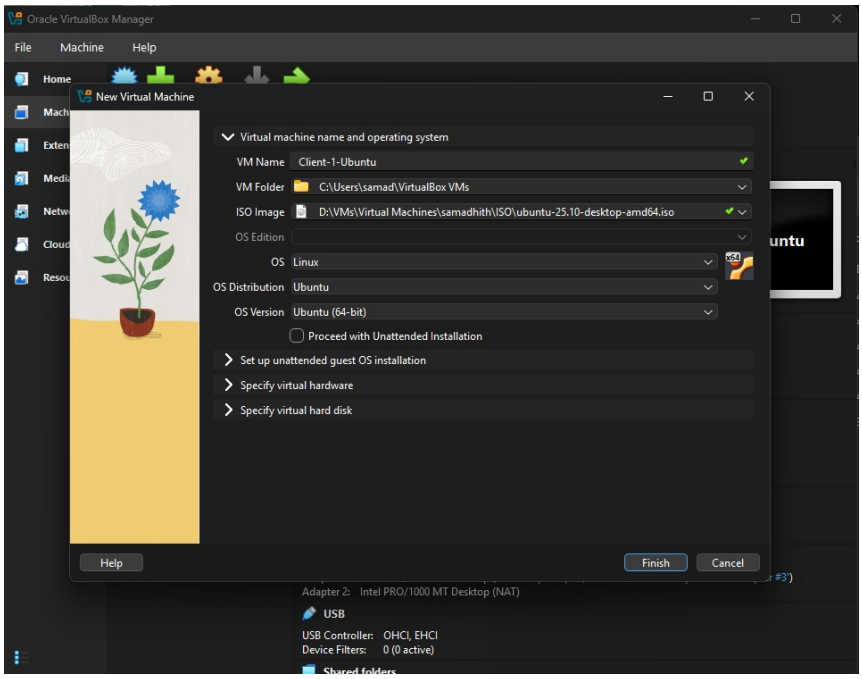


Figure 1.1: Ubuntu Client creation

The Ubuntu client virtual machine was configured using Ubuntu 25.10 desktop version to host the required to test web server access, email communication test and also to serve as an additional test client for domain authentication.

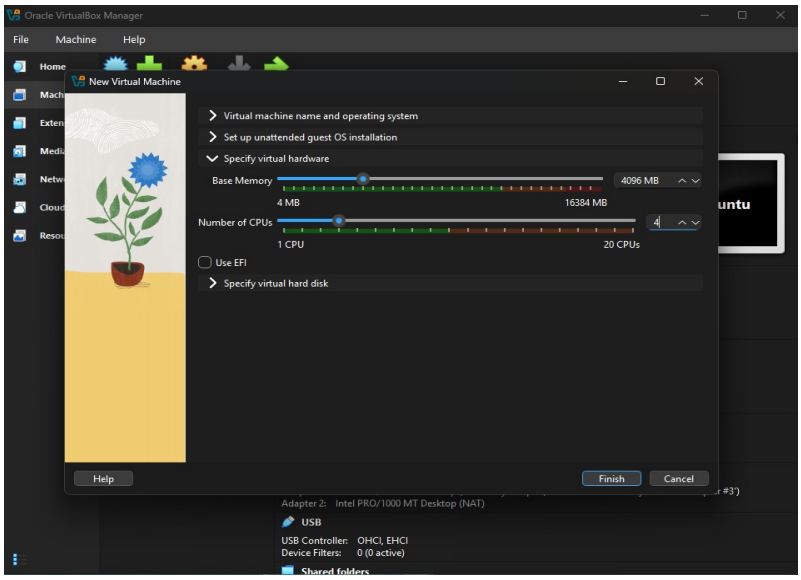


Figure 1.2: Ubuntu Client virtual hardware

The VM was configured with the following hardware specifications:

- **Storage:** 20 GB dynamically allocated virtual disk
- **Memory:** 4 GB RAM allocated to support Zimbra's memory requirements
- **CPU:** 2 vCPUs

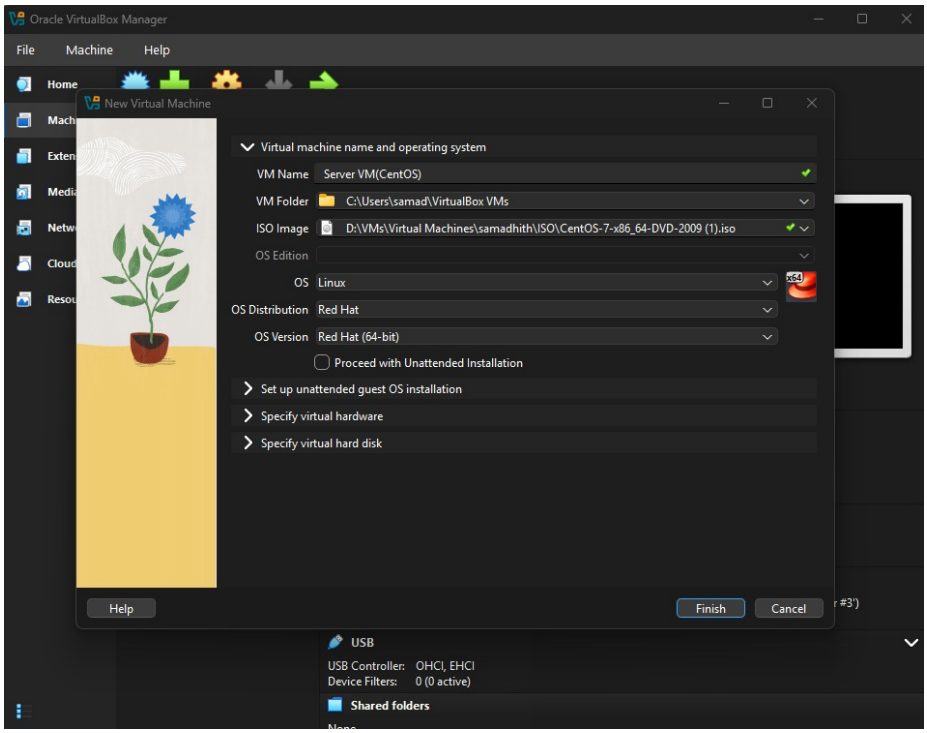


Figure 1.3: CentOS 7 Server creation

The primary CentOS 7 virtual machine was created to host multiple core network services including DHCP server, Zimbra Collaboration Suite mail server and the Operating System is CentOS-7-x86_64-DVD-2009.iso which is most suitable to run Zimbra mail server.

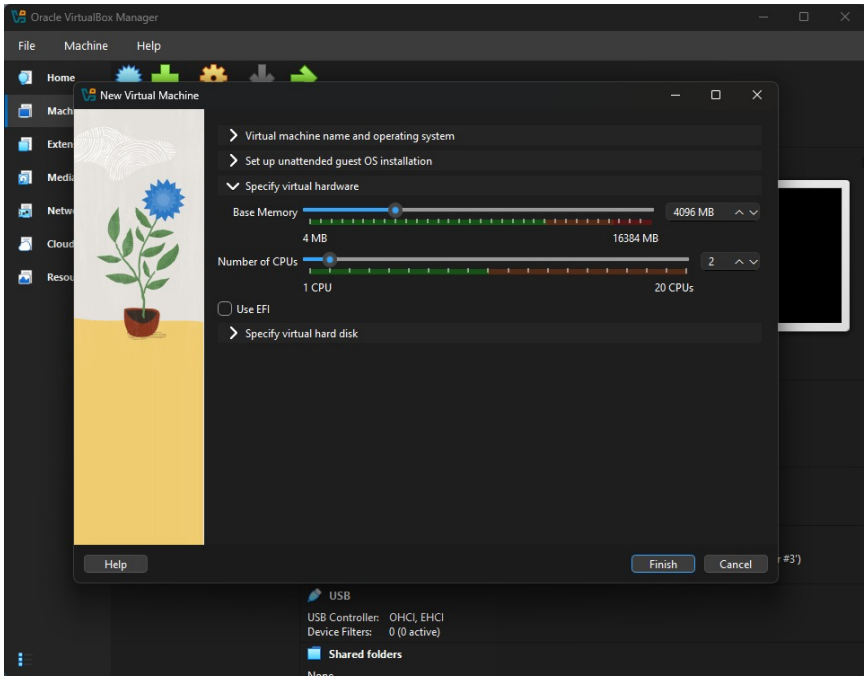


Figure 1.4: CentOS Server virtual hardware

The VM was configured with the following hardware specifications:

- **Storage:** 20 GB dynamically allocated virtual disk
- **Memory:** 4 GB RAM allocated to support Zimbra's memory requirements
- **CPU:** 2 vCPUs

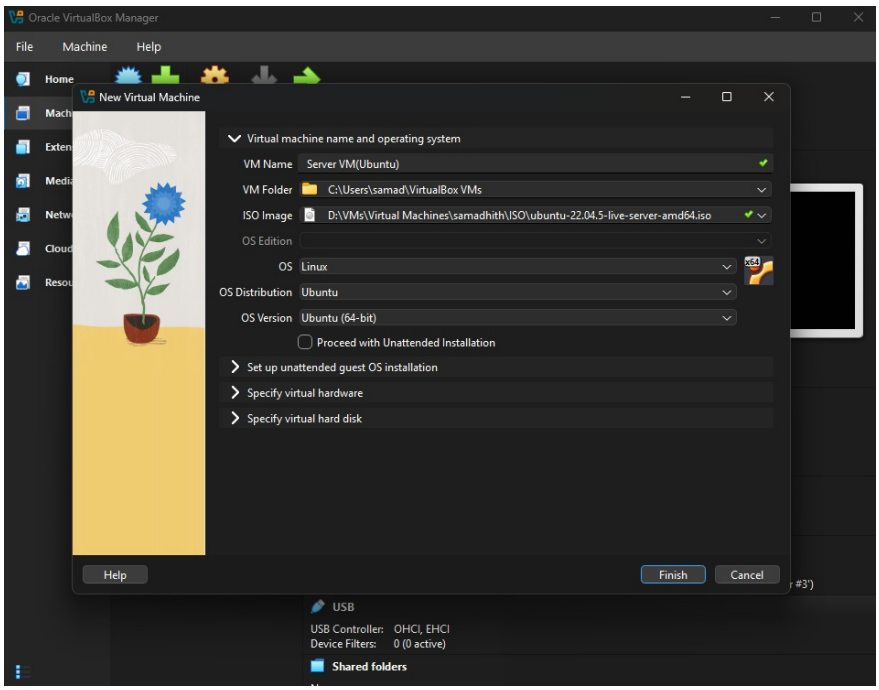


Figure 1.5: Ubuntu Server creation

An additional Ubuntu 22.04 virtual machine was created specifically to host the Samba Active Directory Domain Controller service with Ubuntu 22.04.5 LTS(Server) OS, separating it from the CentOS server to avoid service conflicts.

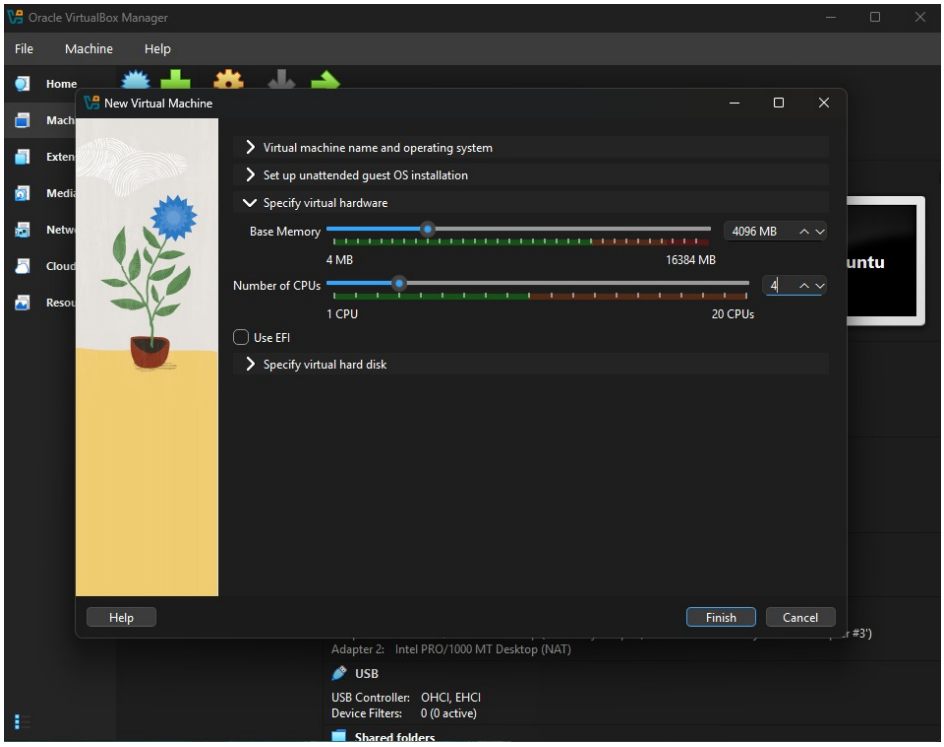


Figure 1.6: Ubuntu Server virtual hardware

The VM was configured with the following hardware specifications:

- **Storage:** 20 GB dynamically allocated virtual disk
- **Memory:** 4 GB RAM allocated to support Zimbra's memory requirements
- **CPU:** 2 vCPUs

b. Operating system installation.

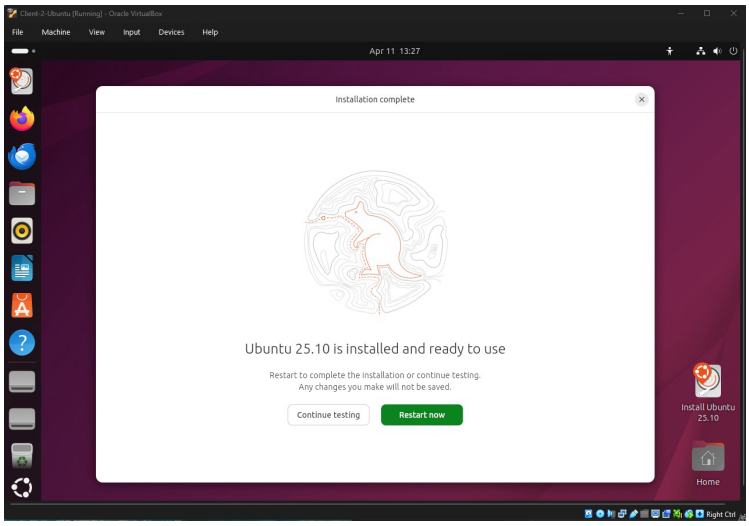


Figure 1.7: Ubuntu Client Installation

Two Ubuntu Desktop 25.10 virtual machines were created to serve as client systems for DHCP testing, domain authentication, and web server access verification. "Normal installation" was selected to include the full desktop environment, web browser, and utility applications required for testing email communication and web page access. After installation, each system was rebooted.



Figure 1.8: CentOS 7 server Installation

This figure shows installation of the CentOS 7 virtual machine using CentOS-7-x86_64-DVD-2009.iso ISO image to provide a lightweight foundation for the DHCP server and Zimbra Collaboration Suite. After configuring the user settings with giving root password and user details, it will starts the installation and after finish that the VM should be reboot.

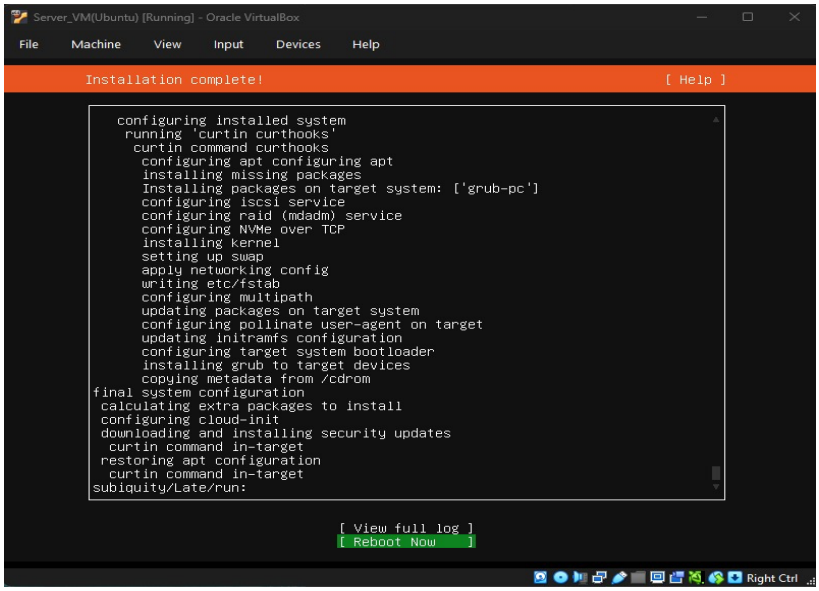


Figure 1.9: CentOS 7 server Installation

In this figure shows an Ubuntu Server 22.04.5 LTS virtual machine was created to host the Samba Active Directory Domain Controller service. "Ubuntu Server" was chosen as the installation type, without any additional snap packages. After the installation completed, the system should be rebooted.

c. Network Configuration.

The laboratory network was constructed using Oracle VM VirtualBox with a combination of Host-Only Adapters and NAT interfaces to enable both internal communication between virtual machines and external internet access.

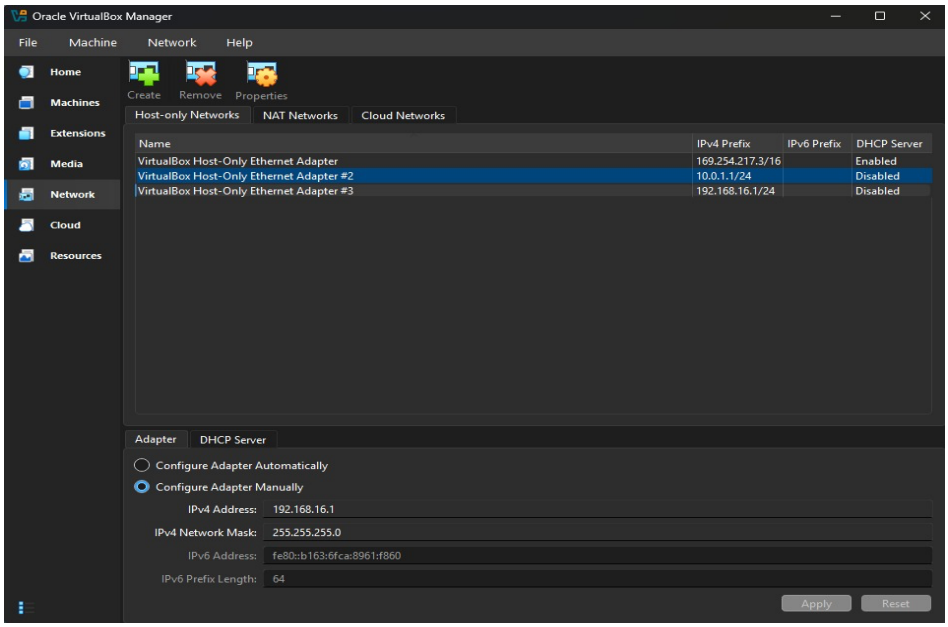


Figure 1.10: Network adapter creation

The figure shows creation of the Virtual-Box Host-Only Adapter #3 for 192.168.16.0/24 private network. The network topology was designed around the assignment-specific subnet 192.168.16.0/24 with the last two digits (16) corresponding to the student's IT number.

- **Adapter Name:** VirtualBox Host-Only Ethernet Adapter #3
- **IPv4 Address:** 192.168.16.1
- **IPv4 Network Mask:** 255.255.255.0
- **DHCP Server:** Disabled (to allow the CentOS server to act as the dedicated DHCP server for the network)

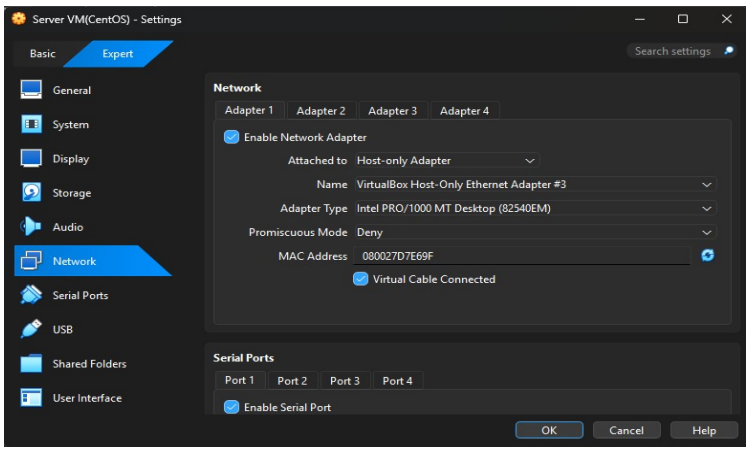


Figure 1.12: Enable network adapter for VM

In this figure it's showing network configuration of the each VM. First select the Virtual Machine and should go to network settings and enable Adapter 1. In there select Host-Only Adapter and select Ethernet Adapter #3. Now that machine enp0s3 interface gets a 192.168.16.0/24 network ip address. Finally Adapter 2 is enabled for NAT (enp0s8).

2. DHCP Server

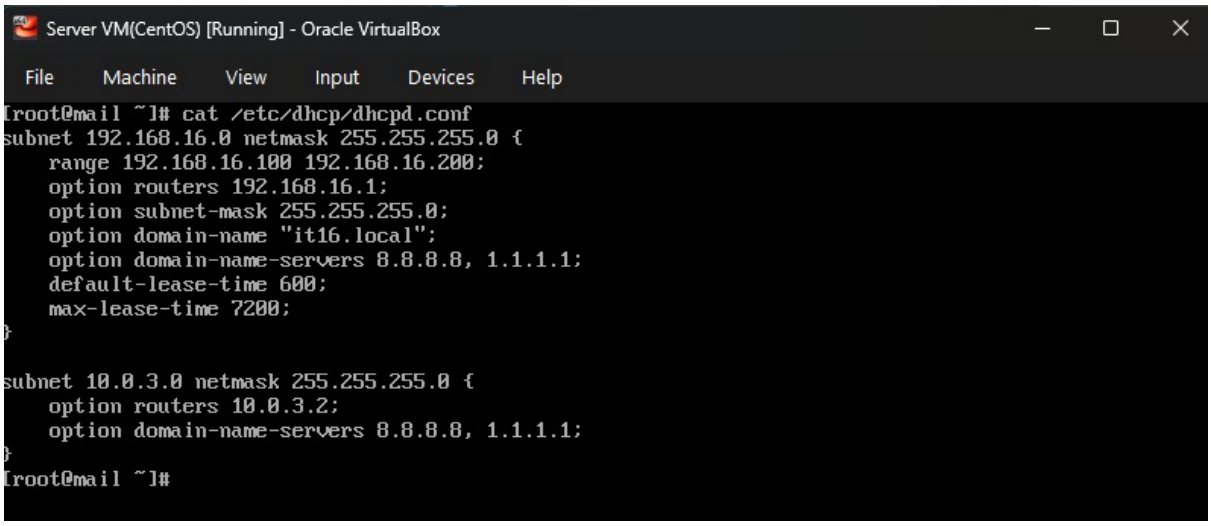


Figure 2.1: DHCP Configuration

This figure presents the DHCP configuration prepared on the CentOS server for the assignment subnet. It verifies that the service was scoped to the required 192.168.16.0/24 network and that the lease pool was restricted to 192.168.16.100 - 192.168.16.200. The configuration also includes the correct subnet mask (255.255.255.0), the default gateway (192.168.16.1), the domain name (it16.local), and external DNS servers (8.8.8.8 and 1.1.1.1). This is important because correct subnet and pool definition is the foundation of controlled dynamic addressing and is one of the core assessed requirements of the assignment. Additionally, the second subnet declaration for the 10.0.3.0/24 NAT interface ensures the CentOS server maintains outbound internet connectivity for package installation and system updates while the client machines remain on the isolated laboratory network.

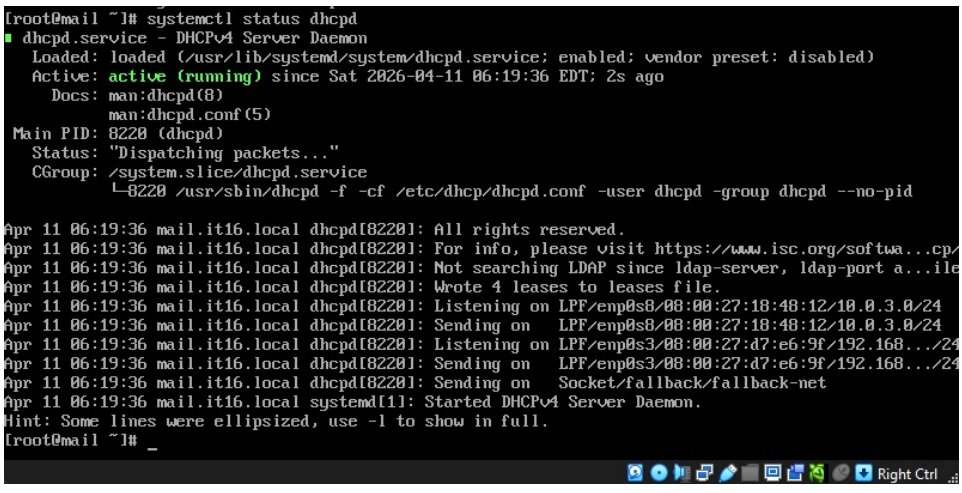


Figure 2.2: DHCP status

This above figure (Figure 2.2) confirms the DHCPv4 Server Daemon is active (running) on the CentOS server, with the service listening on the 192.168.16.0/24 interface as required. The status "Dispatching packets" indicates the server is ready to assign IP addresses to client machines, while the log entry Wrote 4 leases to leases file verifies that previous client assignments have been successfully recorded. This demonstrates that the DHCP service is correctly installed and operational.

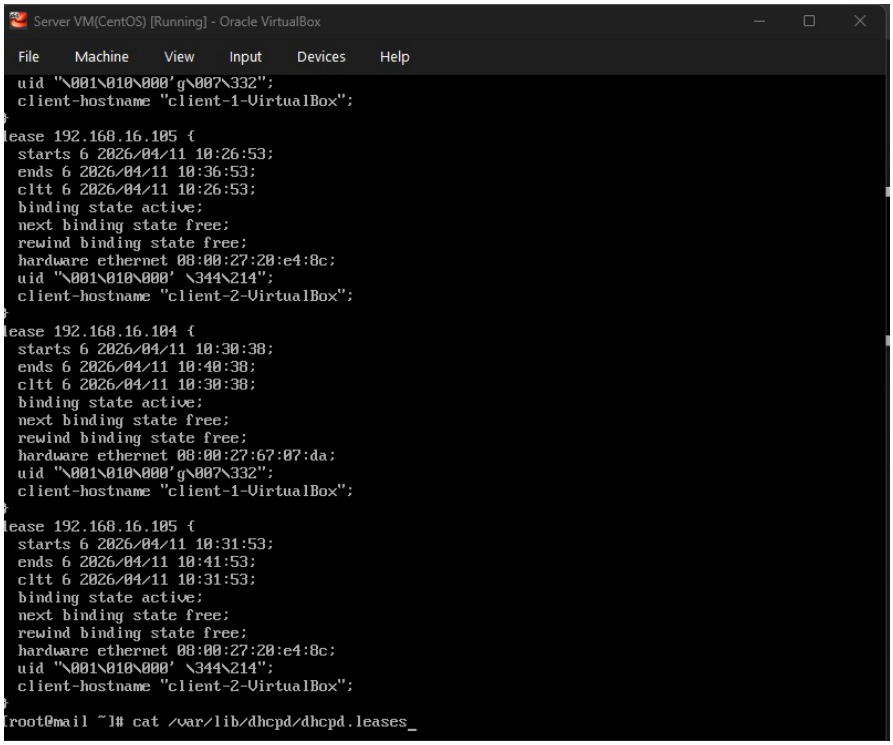


Figure 2.3: Client machines receiving an IP address

This figure displays the contents of the /var/lib/dhcpd/dhcpd.leases file, showing that IP addresses 192.168.16.104 and 192.168.16.105 have been successfully assigned to client machines client-1-VirtualBox and client-2-VirtualBox respectively. The binding state active entries confirm that both clients hold active leases within the required DHCP range (192.168.16.100 - 192.168.16.200). This provides direct evidence that the DHCP server is dynamically allocating IP addresses to client systems as specified in the assignment requirements.

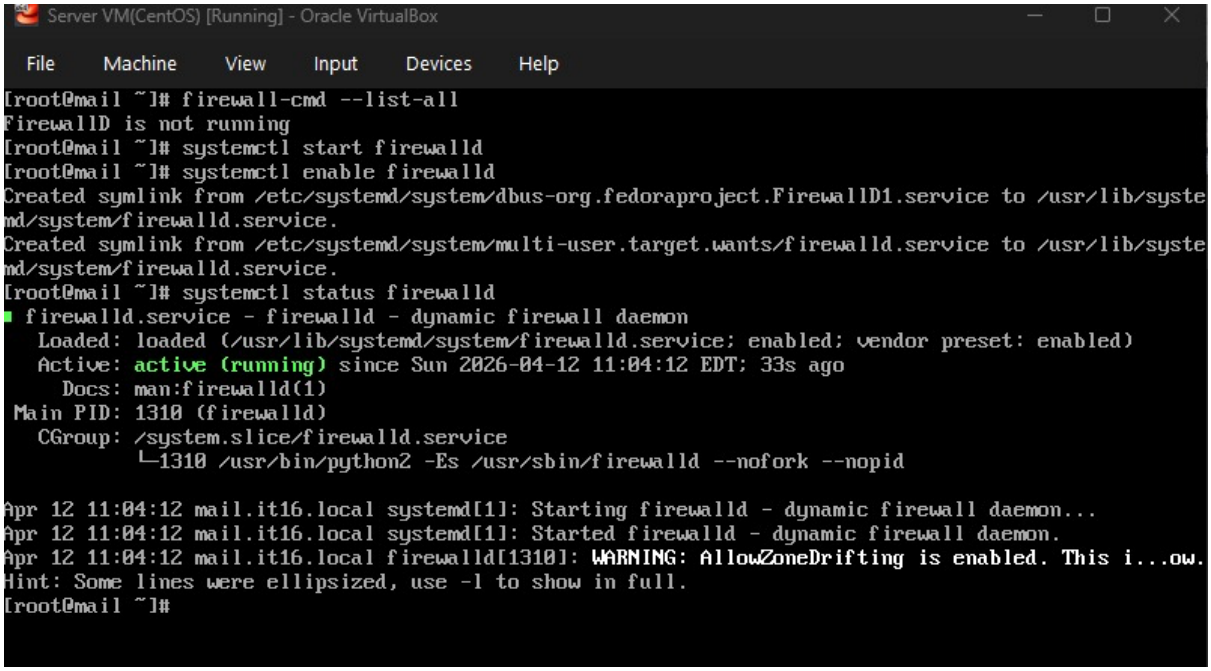


Figure 2.4: Firewall Activation

This figure shows the firewall service (firewalld) being started and enabled on the CentOS server. The first command confirms that the firewall was initially not running. The subsequent commands (systemctl start firewalld and systemctl enable firewalld) start the service immediately and configure it to launch automatically on system boot. The status output verifies that the service is now active (running) with process ID 1310. This is important because a properly configured firewall is essential for securing the network services running on the server, including DHCP, Zimbra mail, and Samba AD, while allowing legitimate client traffic to reach these services as required for the assignment.

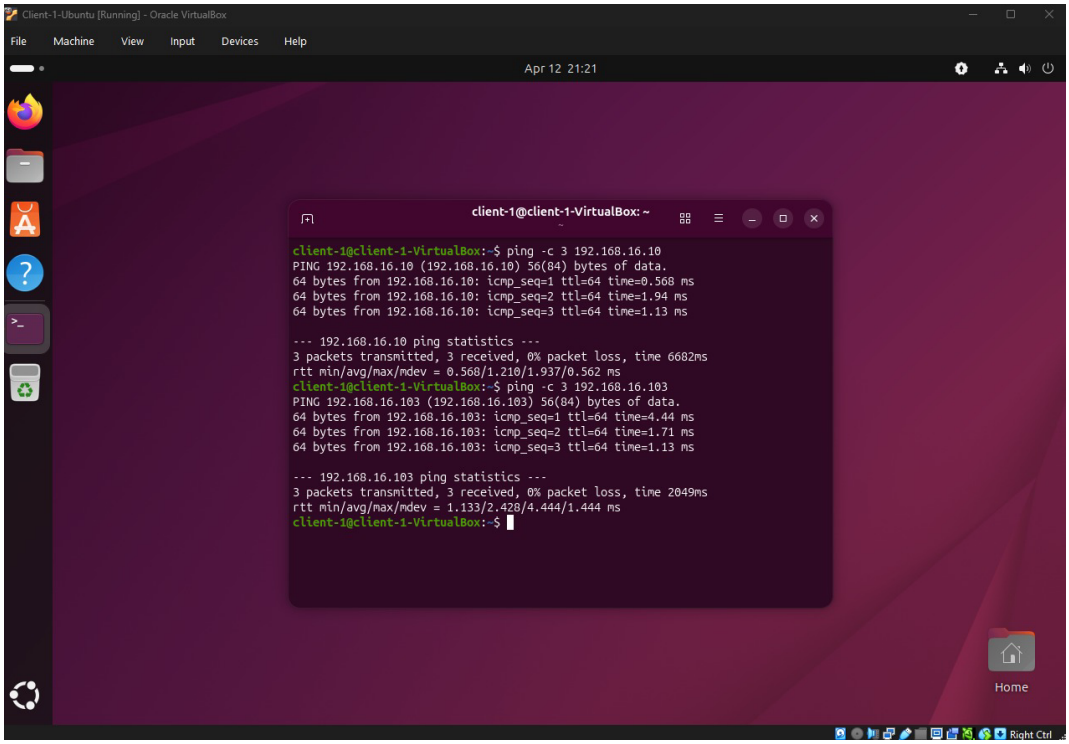


Figure 2.5: Client 1 ping CentOS server (192.168.16.10) and Ubuntu Server (192.168.16.103)

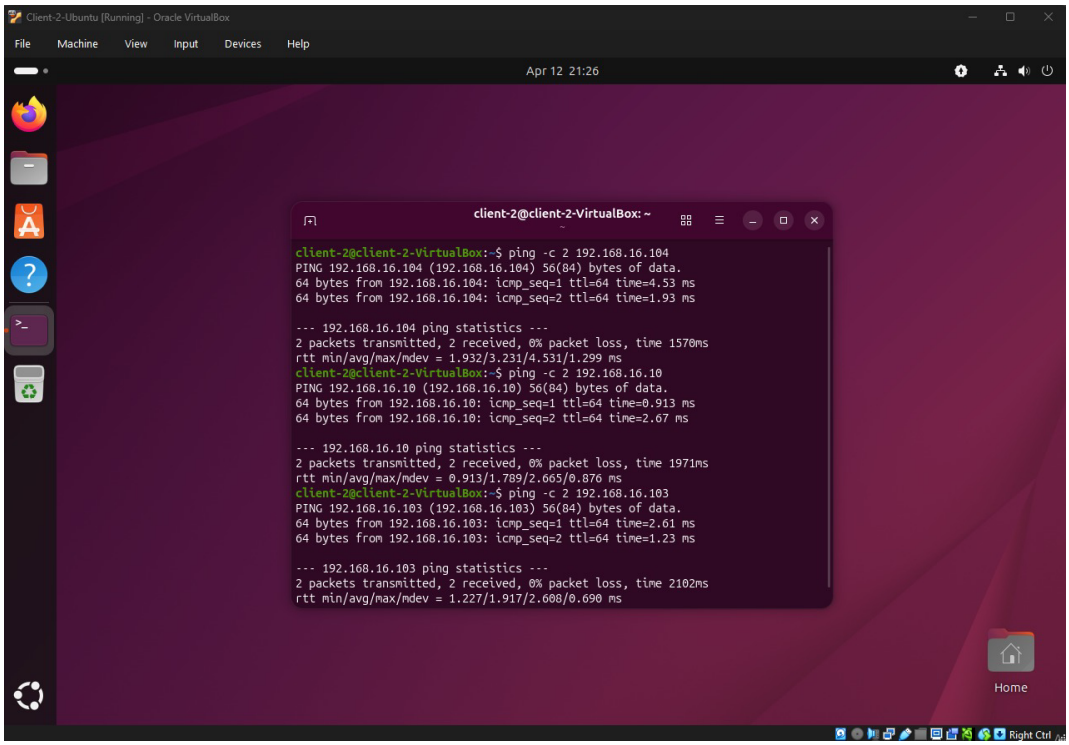


Figure 2.6: Client 2 ping Client1, CentOS server, Ubuntu server.

This figure demonstrates successful network connectivity between all virtual machines in the laboratory environment. The ping commands confirm that Client 1 and Client 2 can communicate with each other, and that both Ubuntu clients can successfully reach the CentOS server. The 0% packet loss results indicate reliable network communication across the 192.168.16.0/24 network. This verification is important because it establishes that the underlying network infrastructure is correctly configured, enabling all dependent services (DHCP, Zimbra mail, Samba AD, and web server) to function properly and communicate as required by the assignment.

3. Zimbra Mail Server

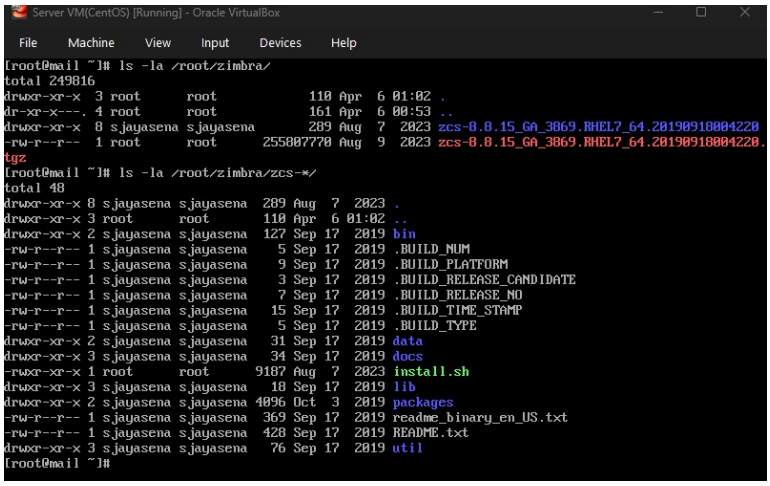


Figure 3.1: Zimbra is downloaded.

The screenshot shows the download of the Zimbra Collaboration Suite using the wget command. The command wget https://files.zimbra.com/downloads/8.8.15_GA/zcs-8.8.15_GA_3869.RHEL7_64_20190917004220.tgz was executed to download the Zimbra installation package from the official Zimbra repository. After the download completed, the command ls -la /root/zimbra/ was used to verify the downloaded file, showing the compressed archive zcs-8.8.15_GA_3869.RHEL7_64_20190917004220.tgz. The command tar -xzvf zcs-.tgz was then run to extract the contents, and ls -la /root/zimbra/zcs-/ confirmed the extracted installation directory containing files such as install.sh, bin, lib, packages, docs, data, and util. This confirms that the Zimbra mail server package has been successfully downloaded and extracted, ready for installation.



Figure 3.2: Zimbra installed and running

This figure displays the output of the zmcontrol status command, confirming that all Zimbra Collaboration Suite services are Running on the host mail.it16.local. The services shown include core components such as mailbox (email storage), mta (mail transfer agent), ldap (directory services), zimbra webapp (webmail interface), and zmconfigd (configuration daemon). This verification is important because it demonstrates that the Zimbra mail server has been successfully installed and fully operational, ready to support email communication between users on the mail.it16.local domain as required by the assignment.

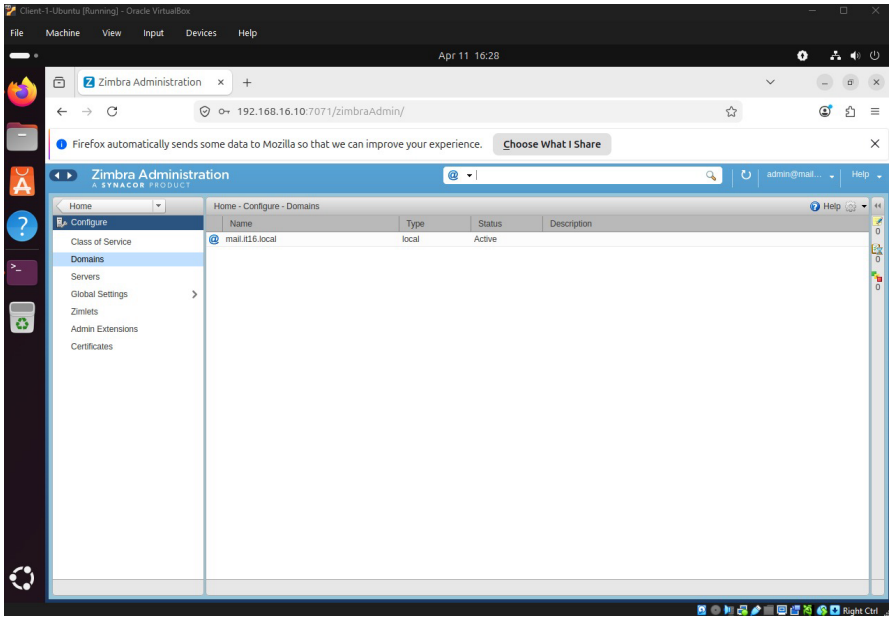


Figure 3.3: Admin Webmail interface

This figure shows the Zimbra Administration Console accessed via https://192.168.16.10:7071, displaying the domain configuration page. The console confirms that the domain mail.it16.local is configured with an Active status and type local, matching the assignment requirement for the mail domain format mail.itXX.local (where XX=16). Accessing the admin console from a client machine web browser demonstrates that the Zimbra server is reachable on the network and that the domain has been successfully created. This verification is important because it confirms both the correct domain configuration and remote accessibility of the Zimbra administration interface, fulfilling the assignment's mail server setup requirements.

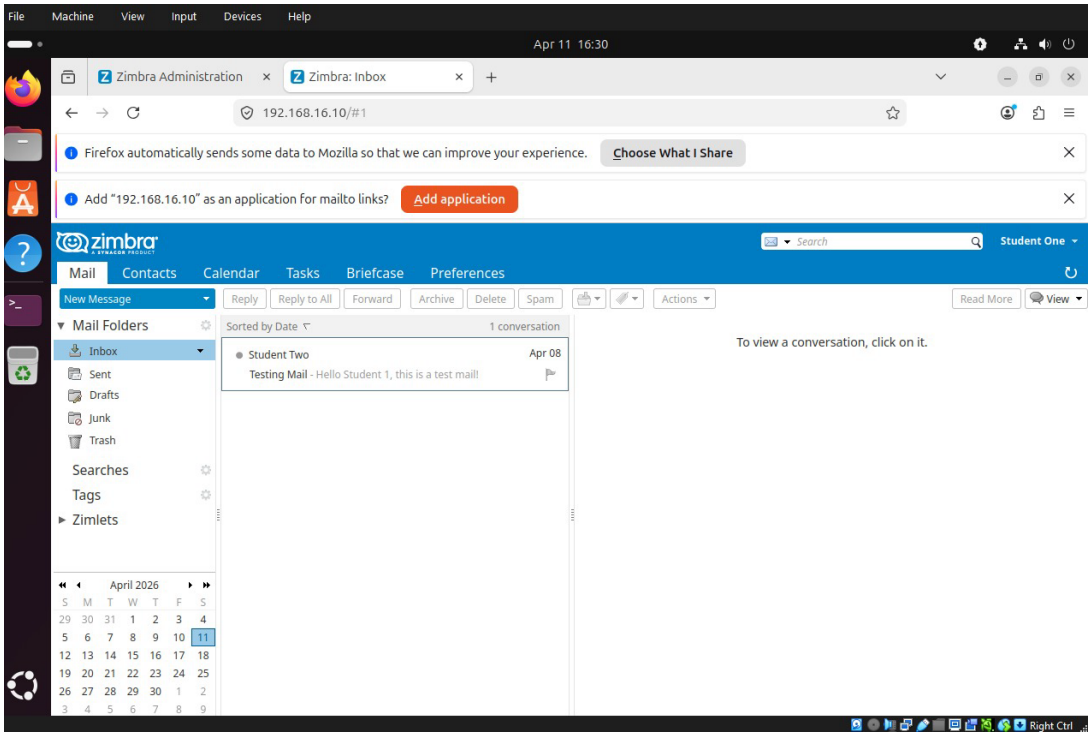


Figure 3.3: Webmail Interface

This figure presents the Zimbra Webmail interface accessed via https://192.168.16.10 from a client machine browser. The login page confirms that the Zimbra mail server is reachable on the network and that the webmail service is operational on the mail.it16.local domain. This accessibility is essential for users to send and receive emails through the web interface. Once logged in, users can compose messages, manage their inbox, and communicate with other users on the same domain. This verification demonstrates that the Zimbra mail server has been successfully deployed and that the webmail interface is properly accessible from client systems, fulfilling the assignment requirement for email communication demonstration.

4. Samba Active Directory Domain Controller

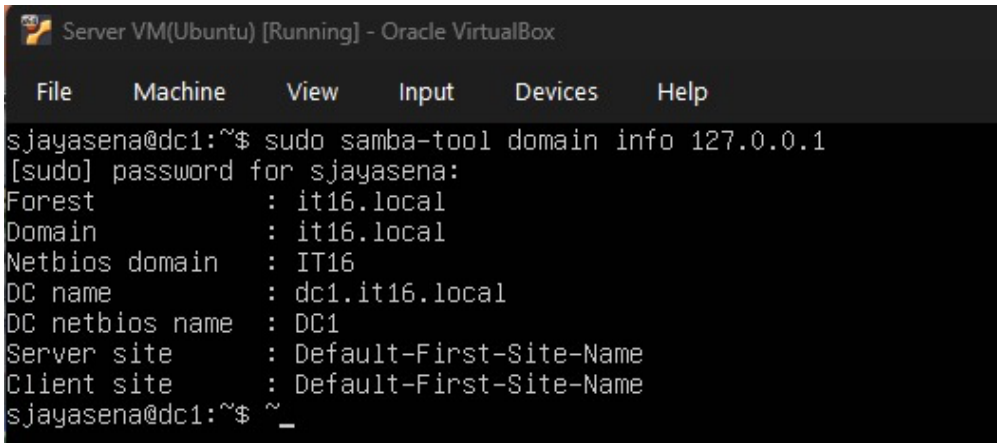


Figure 4.1: Domain creation

The figure demonstrates the successful creation of the Samba Active Directory Domain Controller. The command sudo samba-tool domain info 127.0.0.1 confirms that the domain it16.local has been properly provisioned, with the forest name it16.local, NetBIOS domain name IT16, and the domain controller identified as dc1.it16.local. This verifies that the Samba AD DC is functioning correctly.

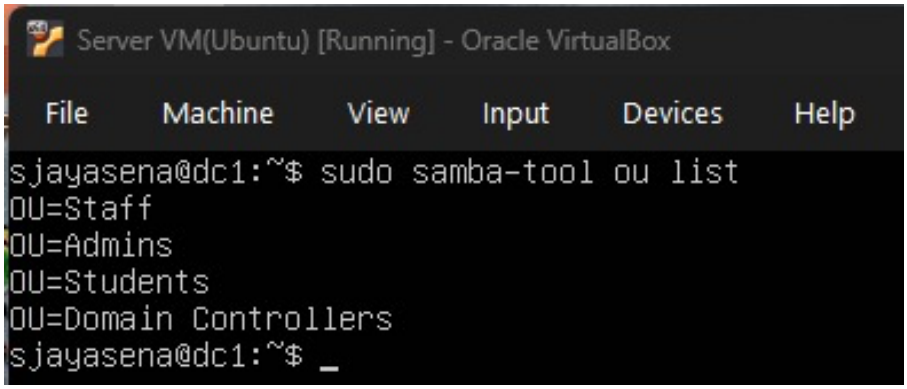


Figure 4.2: Organizational Units

The screenshot shows the Organizational Units (OUs) created in the Samba Active Directory Domain Controller. The command sudo samba-tool ou list displays the OUs: OU=Admins, OU=Staff, and OU=Students, along with the default OU=Domain Controllers This confirms that the required OUs have been successfully created according to the assignment specifications.

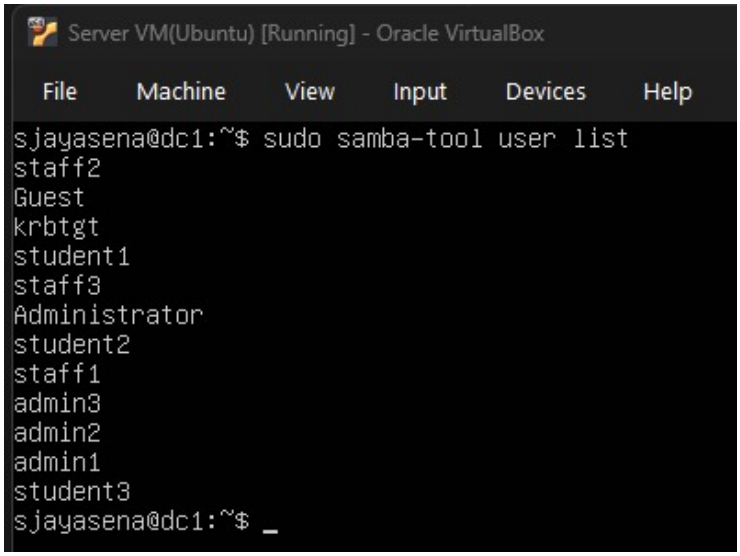


Figure 4.3: Created User accounts

The screenshot displays the list of domain user accounts created in the Samba Active Directory Domain Controller. The command sudo samba-tool user list was executed to view all existing users in the domain it16.local. The output shows the three administrator users admin1, admin2, and admin3, the three staff users staff1, staff2, and staff3, and the three student users student1, student2, and student3. Additionally, the default domain accounts such as Administrator, Guest, and krbtgt are also visible. This confirms that all nine required user accounts have been successfully created according to the naming schema specified in the assignment.

5. Domain Client Integration

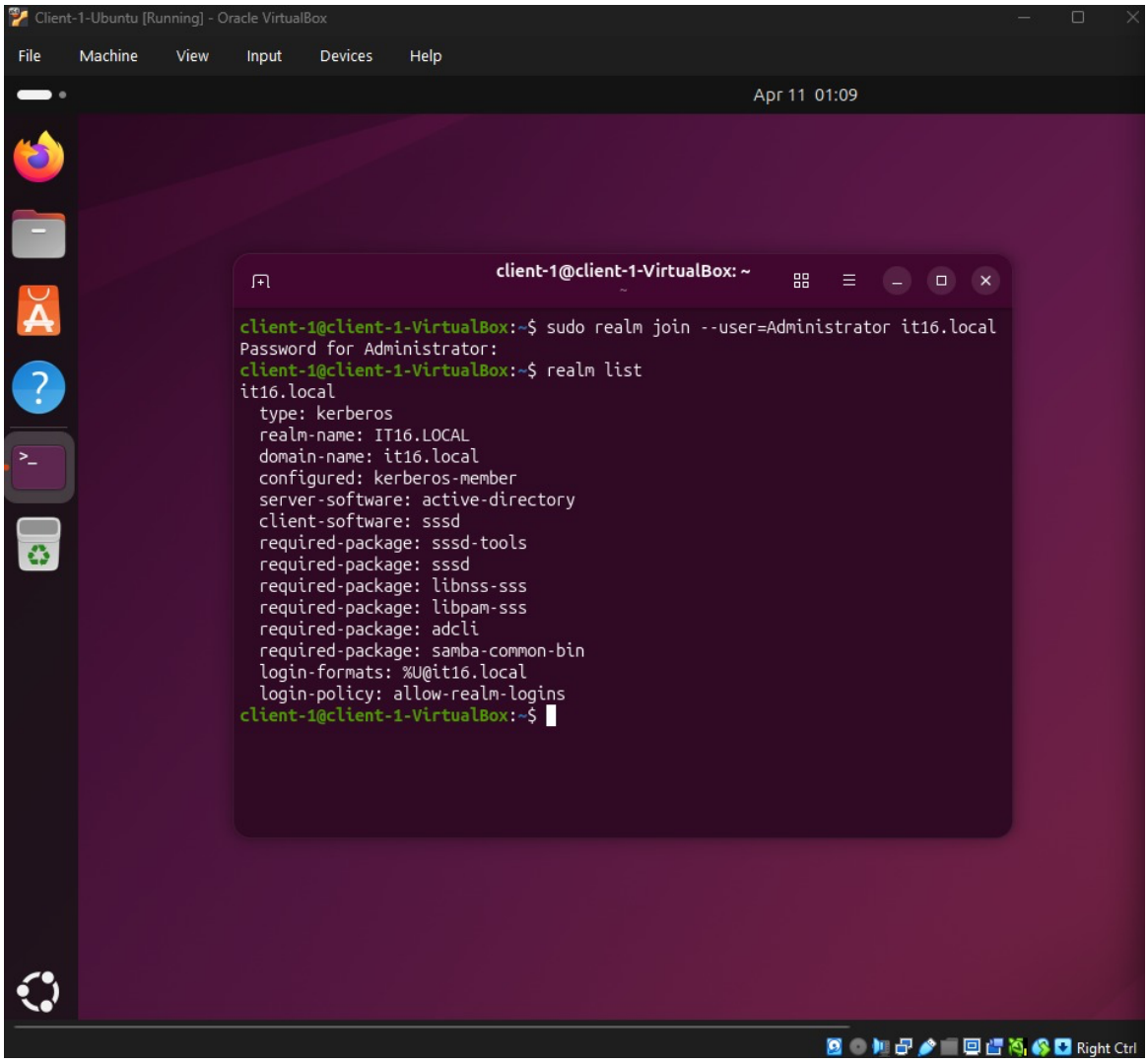


Figure 5.1: Client machine joined to the domain

The figure demonstrates the successful integration of the Ubuntu client machine into the Samba Active Directory domain. First, the command sudo realm join --user=Administrator it16.local was executed, which authenticated the client to the domain controller using the Administrator account. Following the successful join, the command realm list it16.local was run to verify the domain membership status. The output confirms that the domain it16.local is configured as a kerberos-member with server-software set to active-directory and client-software set to sssd. The required packages for domain integration are listed, and the login format is specified as username@it16.local. This confirms that the client machine has successfully joined the domain it16.local and is ready to authenticate domain users.

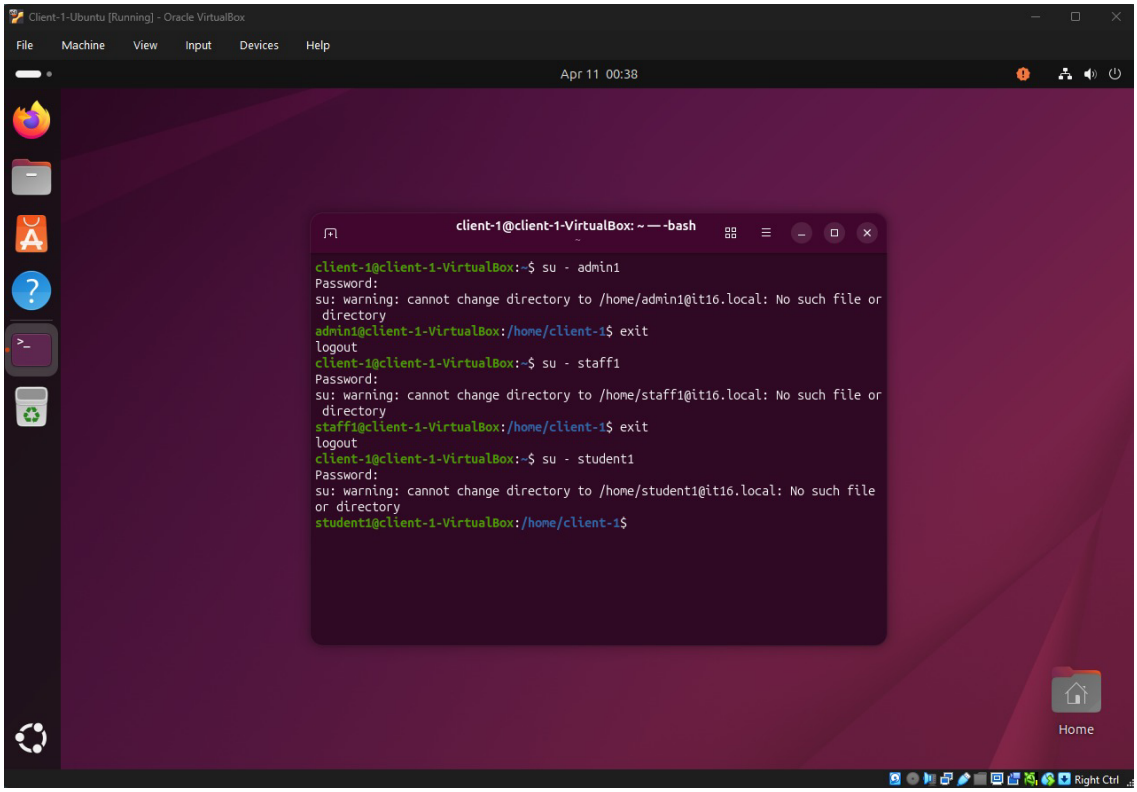


Figure 5.2: Logging using domain user account

The figure demonstrates successful domain authentication by logging into the client machine using domain user accounts from each organizational unit. The commands su - admin1, su - staff1, and su - student1 were executed on the client machine, each prompting for a password and successfully authenticating the respective domain user. After each successful login, the terminal prompt changed from client-1@client-1-VirtualBox to admin1@client-1-VirtualBox, staff1@client-1-VirtualBox, and student1@client-1-VirtualBox respectively. This confirms that the domain users admin1, staff1, and student1 have successfully authenticated against the Samba Active Directory Domain Controller running on it16.local. The warning messages about missing home directories do not affect the authentication process, as the users are still able to log in and the identity change is clearly visible from the updated terminal prompt. This verifies that domain authentication is functioning correctly for all three user levels as required by the assignment.

6. Web Server

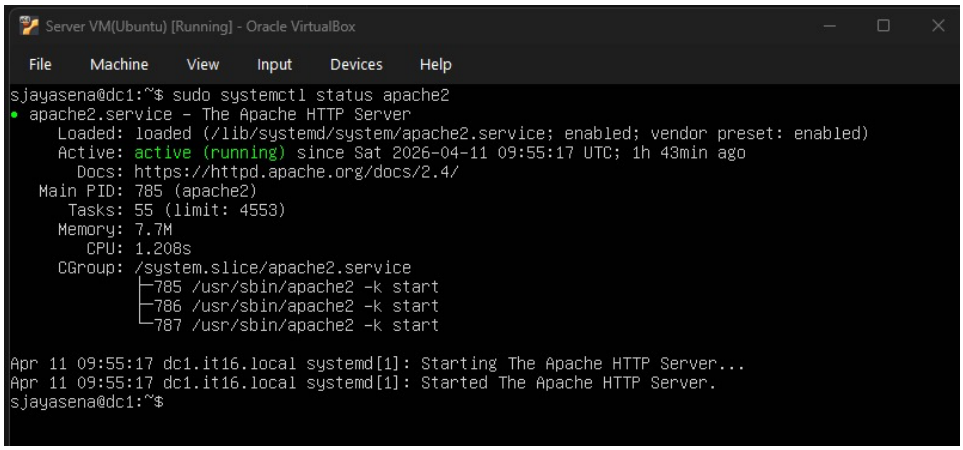


Figure 6.1: Web server is running

The figure shows the Apache HTTP server status on the Ubuntu web server, confirming that the service is loaded, enabled to start on boot, and has been actively running for 1 hour and 43 minutes since April 11, 2026 at 09:55:17 UTC. The display includes the main process ID 785, three Apache child processes, memory usage of 7.7 MB, and system log entries indicating that the server started successfully without errors. This verifies that the web server is properly installed and operational on the Ubuntu machine.

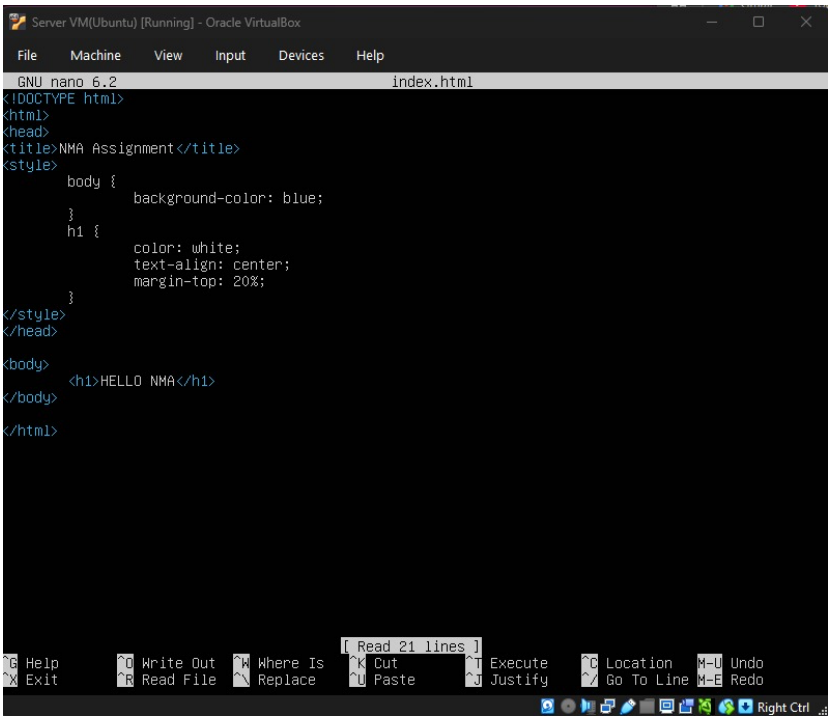


Figure 6.2: Web page source configured to display HELLO NMA with a blue background.

The figure displays the HTML code for the HELLO NMA webpage opened in the GNU nano text editor on the Ubuntu server. The code includes a DOCTYPE declaration and HTML structure with a head section containing a title and style rules. The style section defines a blue background color for the body element and white, centered text with 20 percent top margin for the h1 heading. The body section contains the h1 heading with the text HELLO NMA. The nano editor interface shows the file name index.html at the top and indicates that 21 lines have been read into the editor, confirming that the webpage has been successfully created and saved in the web server directory.

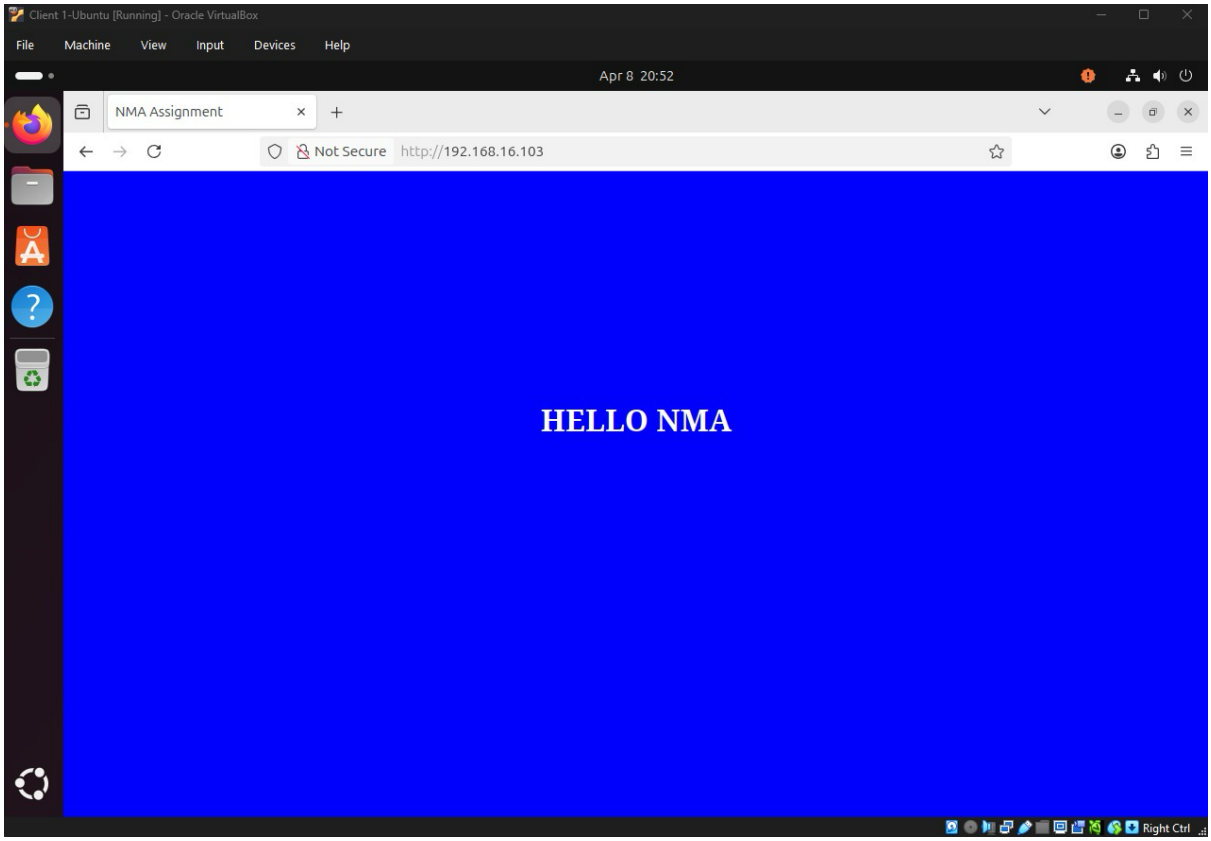


Figure 6.2: Web page is displaying and accessed from client machine

The figure captures the client machine accessing the web server, where typing the URL http://192.168.16.103 in the web browser successfully displays the webpage. The browser window shows the page with a blue background and the text HELLO NMA clearly visible in the center. The address bar confirms that the connection is made to the server at the specified IP address using standard HTTP on port 80. This demonstrates that the web server is correctly configured and accessible from the client machine, fulfilling the requirement for remote access to the hosted webpage.

7. Verification

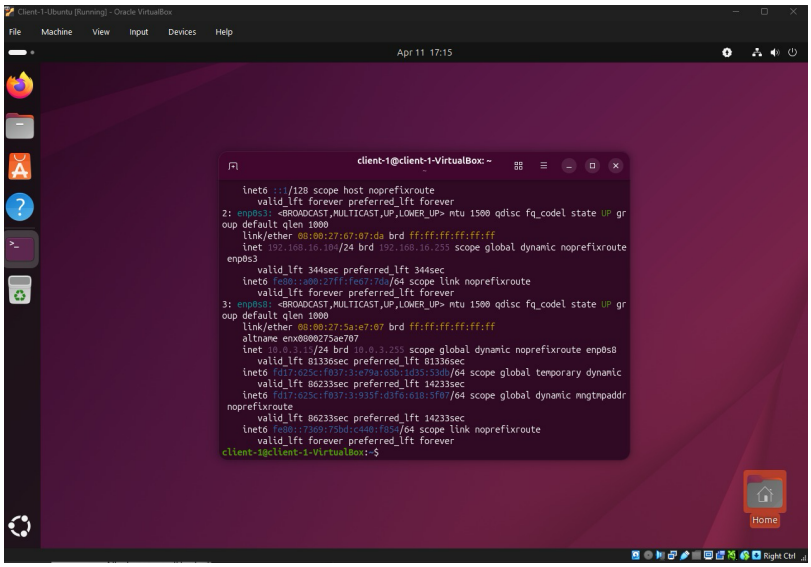


Figure 7.1: Client machine receives an IP address from the DHCP server

This figure displays the output of the ip a command on an Ubuntu client machine, confirming successful IP address assignment from the CentOS DHCP server. The client interface enp0s3 shows an IPv4 address of 192.168.16.104/24 with the dynamic flag, indicating the address was automatically obtained via DHCP. The valid_lft 344sec entry confirms an active lease duration. This address falls within the required DHCP range of 192.168.16.100 to 192.168.16.200, demonstrating that the client machine has successfully received an IP address automatically as specified in the assignment requirements. The second interface enp0s8 shows a NAT address (10.0.3.15) for internet connectivity.

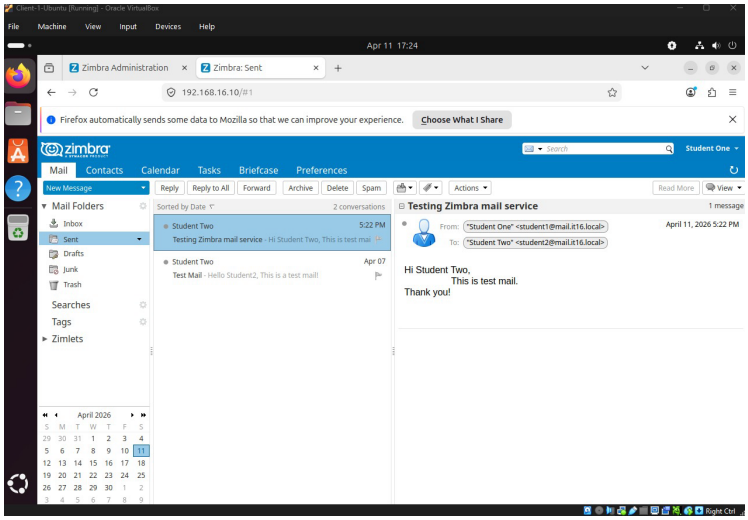


Figure 7.2: Student One is sending a mail to Student Two using Zimbra mail service

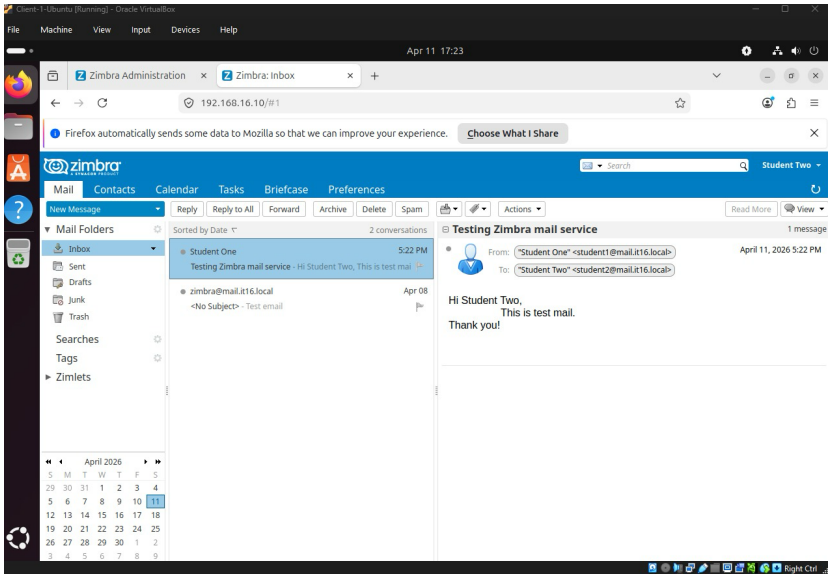


Figure 7.3: Student Two receives the mail is sent from the Student One.

These figures demonstrate successful email communication between users on the Zimbra mail server. The first screenshot shows Student One logged into the Zimbra webmail interface at https://192.168.16.10, composing a new message addressed to student2@mail.it16.local with a subject and body content. The second screenshot shows Student Two's inbox, displaying the received email from Student One, confirming successful delivery. Together, these screenshots provide clear evidence that the Zimbra mail server is fully operational, that user accounts have been correctly created with a proper naming schema (student1 and student2), and that email messages can be both sent and received between users on the mail.it16.local domain. This verification fulfills the core assessed requirement.

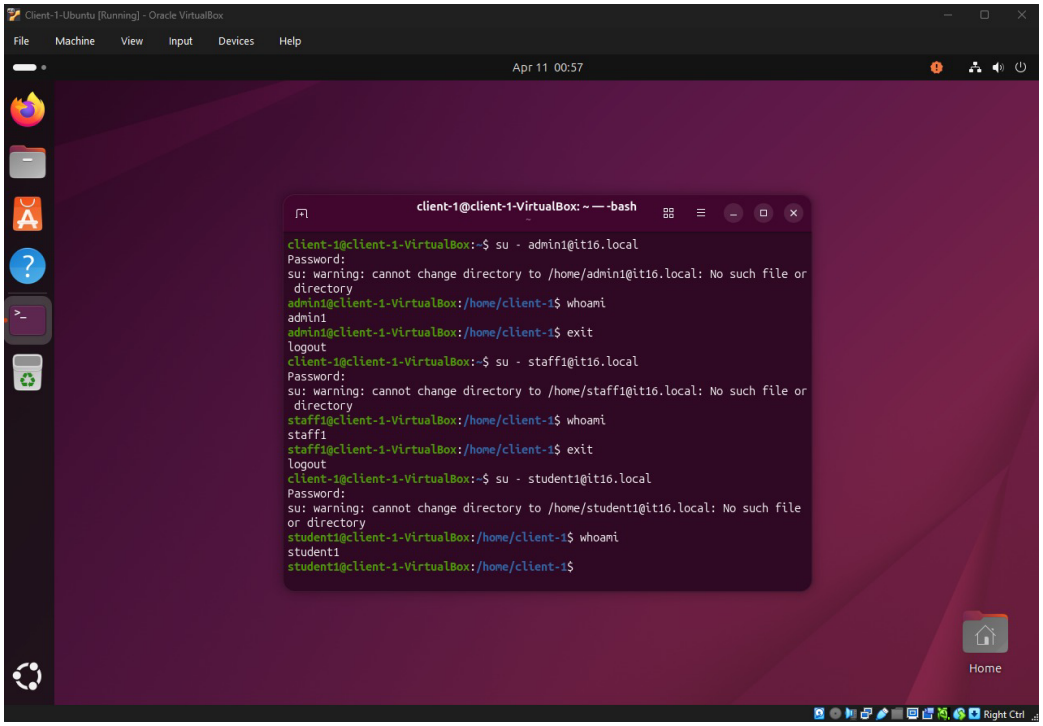


Figure 7.4: Domain authentication for admin1, staff1, student1

The screenshot demonstrates successful domain authentication by verifying that domain user accounts can log into the client machine. The commands su - admin1@it16.local, su - staff1@it16.local, and su - student1@it16.local were executed, each prompting for a password and successfully authenticating the respective domain user. After each login, the whoami command was run, confirming that the active session belongs to admin1, staff1, and student1 respectively. Although warning messages about missing home directories appear, this does not affect the authentication process, as the users are successfully logged in and their identities are correctly displayed. This confirms that all three domain user levels can authenticate against the Samba Active Directory Domain Controller as required by the assignment.

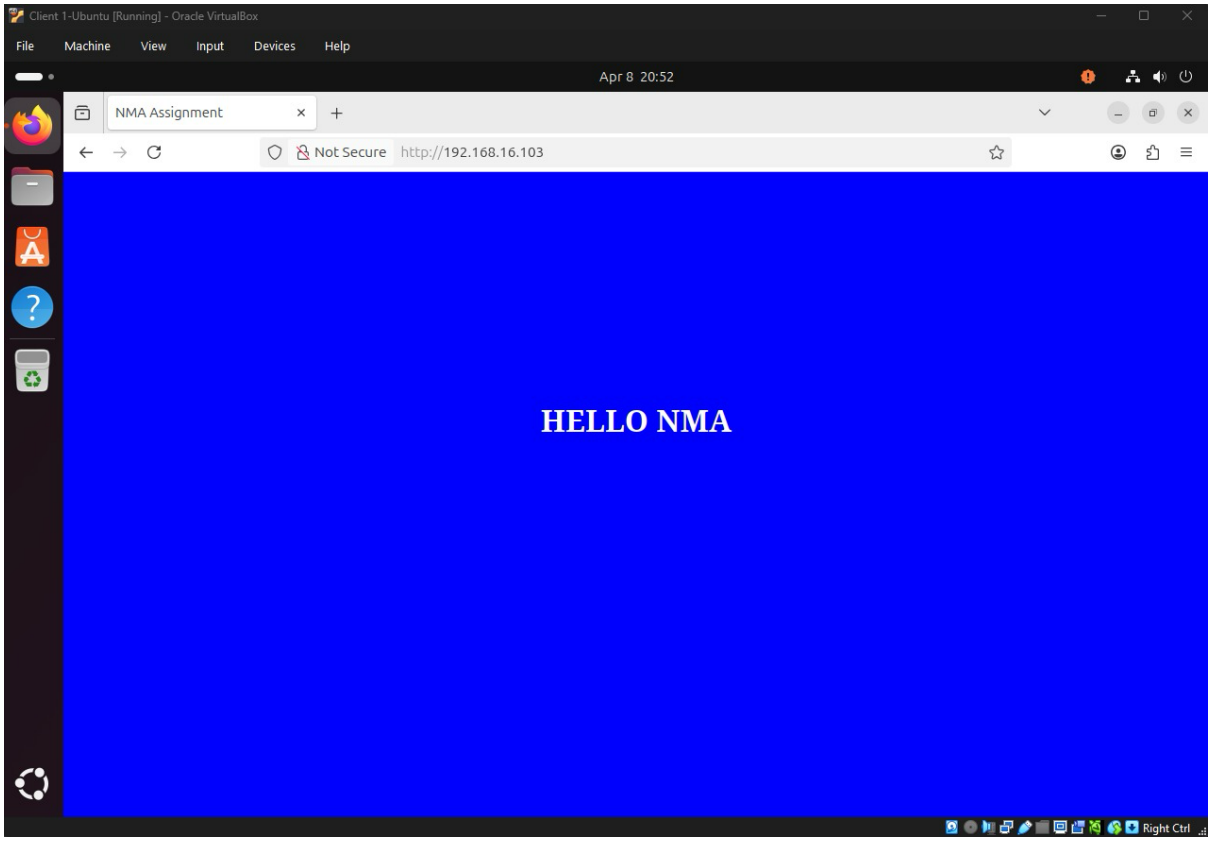


Figure 7.5: Client machines have the web server access.

The final verification screenshot shows the client machine successfully accessing the HELLO NMA webpage through a web browser. The browser address bar displays the URL http://192.168.16.103, confirming that the client is connecting directly to the Ubuntu web server using its IP address. The webpage renders fully with a solid blue background covering the entire browser window, and the text HELLO NMA appears prominently in white color at the center of the page. This demonstrates that the Apache web server is correctly configured, the HTML file with the blue background styling is properly placed in the web root directory, the network connectivity between the client and server is functional, and the firewall allows HTTP traffic on port 80. The successful display of the webpage from the client machine verifies that all web server requirements have been met and the service is fully operational.

8. Conclusion

This assignment successfully deployed and validated a Linux-based network environment delivering the required core administrative services within a virtual laboratory. The implementation fulfilled all specified requirements across the five main components.

For Part A (Initial Setup), three virtual machines were successfully created and configured with CentOS 7 as the primary server and Ubuntu 22.04/25.10 as client systems. Network connectivity was verified between all machines using ping commands, establishing a stable foundation for subsequent services.

For Part B (DHCP Server), the CentOS server was configured as a dedicated DHCP server with the address range 192.168.16.100 - 192.168.16.200 on the 192.168.16.0/24 network, matching the assignment specification (X=16). Both Ubuntu client machines successfully received IP addresses automatically within the configured range, as confirmed by the DHCP leases file and client IP address verification.

For Part C (Zimbra Mail Server), Zimbra Collaboration Suite was successfully installed and configured on the CentOS server with the mail domain mail.it16.local. Multiple user accounts (student1, student2, staff1) were created following a proper naming schema. Email communication was successfully demonstrated between users using the Zimbra web interface accessed via https://192.168.16.10, with both sending and receiving functionality confirmed.

For Part D (Samba Active Directory Domain Controller), Samba was configured as an AD DC for the it16.local domain. The required Organizational Units (Admins, Staff, Students) were created, with three user accounts provisioned within each OU. Domain authentication was successfully demonstrated by joining an Ubuntu client to the domain.

For Part E (Web Server), an Apache web server was installed and configured on the Ubuntu client machine, publishing a webpage displaying "HELLO NMA" with a blue background. The webpage was successfully accessed from client machines using a web browser, confirming proper web server functionality.

All services were verified to be functioning correctly, including DHCP IP allocation, email communication, domain authentication, and web server access. The completed environment meets all learning outcomes and assessment criteria specified in the assignment brief. Any platform inconsistencies noted in the requirements were resolved through careful service placement, ensuring a fully operational and demonstrable network infrastructure.

9. References

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